MCQ 1: Understand and Choose

Five teaching questions with one answer. Choose one answer per question; then explain one
This chapter format does not reproduce the official format of the EAM.

1. $u_n = 3n + 2$: u_4 value
   - A. 14
   - B. 18
   - C. 12
   - D. 6
2. $u_0 = 2$, $u_{n+1} = u_n + 5$: u_3 value
   - A. 10
   - B. 12
   - C. 17
   - D. 15
3. An 8 per cent increase is the common difference / ratio
   - A. 0,08
   - B. 1,8
   - C. 0,92
   - D. 1,08
4. From u_3 to u_9 included, we count
   - A. 6
   - B. 7
   - C. 9
   - D. 12
5. $1 + 2 + \cdots +$ 20 is
   - A. 200
   - B. 420
   - C. 210
   - D. 190

MCQ 2: understand and choose

Five teaching questions with one answer. Choose one answer per question; then explain one
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1. $u_n = -3 \times 2^n$ is
   - A. Strictly increasing
   - B. constant
   - C. Neither increasing nor decreasing
   - D. Strictly decreasing
2. 1 + 2 + 4 + 8 is
   - A. 15
   - B. 16
   - C. 14

D. 8

3. For $u_n = \dfrac{16}{2^n}$, first row where $u_n < 2$:

A. 2

B. 3

C. 4

D. 5

4. The future $(-1)^n$

A. tend towards 0

B. has no limit

C. is increasing

D. is constant

5. A list contains u_0 to u_{10} ; its length is

A. 10

B. 9

C. 12

D. 11

True or false?

1. Any geometric sequence of common ratio $q > 1$ is increasing.

2. If two sequences have their first three equal terms, they are equal.

3. For $q = 1$, $1 + q + \cdots + q^n = n + 1$.

Personal review

I distinguish explicit and recurring; I recognize a constant increase or rate; I count the terms; I control the previous rank of a threshold.
For each objective, indicate: acquired without help; succeeded with help; to rework. Write an example and a mistake to avoid.